Computer Graphics

Graphics Primitives & Transformations

Lines and Polygons

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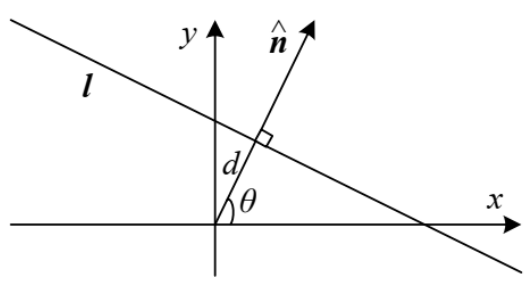
2D Line

- 2D line in homogeneous coordinates system $\tilde{\mathbf{l}} = (a, b, c)$
- Equation of line $\bar{\mathbf{x}} \cdot \tilde{\mathbf{l}} = ax + by + c = 0.$

$$\mathbf{l} = (\hat{n}_x, \hat{n}_y, d) = (\hat{\mathbf{n}}, d) \text{ with } \|\hat{\mathbf{n}}\| = 1$$

$\hat{\mathbf{n}}$ is the normal vector
d is the distance to origin



$$\hat{\mathbf{n}} = (\hat{n}_x, \hat{n}_y) = (cos\theta; sin\theta)$$

(θ, d) is polar coordinates

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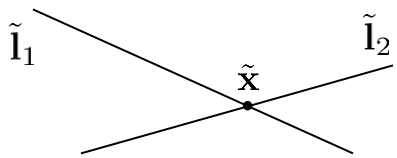
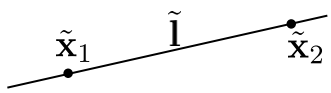
Why Homogeneous Coordinates

- Intersection of two lines as $\tilde{\mathbf{x}} = \tilde{\mathbf{l}}_1 \times \tilde{\mathbf{l}}_2$



- Line joining two points as $\tilde{\mathbf{l}} = \tilde{\mathbf{x}}_1 \times \tilde{\mathbf{x}}_2$



Find the intersection point of

$$x + 2y + 1 = 0$$
$$2x + 3y + 5 = 0$$

Point of intersection $\mathbf{x} = \begin{bmatrix} -7 \\ 3 \end{bmatrix}$

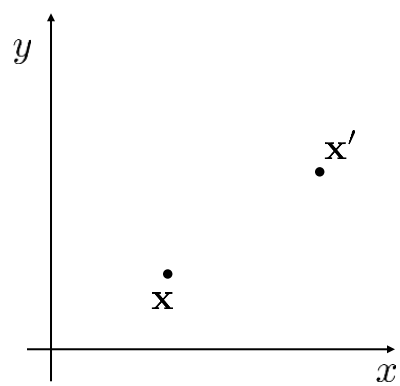
2D Translations

2D translation are denoted by

$$\mathbf{x}' = \mathbf{x} + \mathbf{t}$$
$$= \begin{bmatrix} \mathbf{I} & \mathbf{t} \end{bmatrix} \bar{\mathbf{x}}$$

$\mathbf{I}$: 2×2 Identity Matrix
$\mathbf{t}$: Translation Vector

$$\bar{\mathbf{x}}' = \begin{bmatrix} \mathbf{I} & \mathbf{t} \\ \mathbf{0}^T & 1 \end{bmatrix} \bar{\mathbf{x}}$$

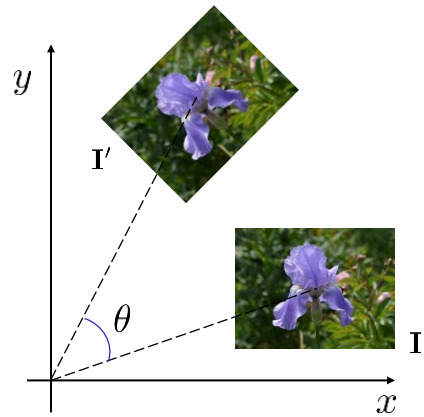


2D Euclidean Transformation

2D Rigid body motion

$$\mathbf{x}' = \mathbf{R}\mathbf{x} + \mathbf{t}$$
$$= \begin{bmatrix} \mathbf{R} & \mathbf{t} \end{bmatrix} \bar{\mathbf{x}}$$

where $\mathbf{R} = \begin{bmatrix} cos(\theta) & -sin(\theta) \\ sin(\theta) & cos(\theta) \end{bmatrix}$



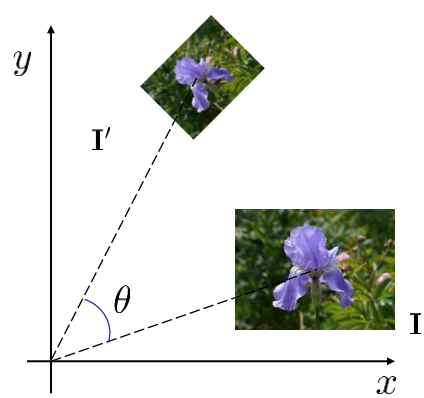
is an orthonormal rotation matrix $\mathbf{R}\mathbf{R}^T = \mathbf{I}$ and $|\mathbf{R}| = 1$

2D Scaled Rotation

2D Similarity Transform

$$\mathbf{x}' = s\mathbf{R}\mathbf{x} + \mathbf{t}$$
$$= \begin{bmatrix} s\mathbf{R} & \mathbf{t} \end{bmatrix} \bar{\mathbf{x}}$$
$$= \begin{bmatrix} a & -b & t_x \\ b & a & t_y \end{bmatrix} \bar{\mathbf{x}}$$

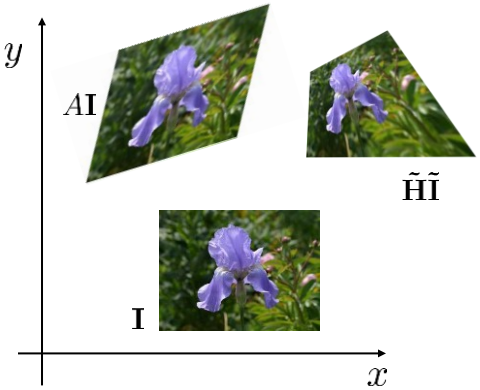
s is an arbitary scaling factor



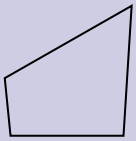
2D Affine and Projective

2D Affine Transform, $\mathbf{x}' = A\mathbf{x}$
where A is an arbitary 2×3 matrix

Projective/Perspective Transform
Homography
$\tilde{\mathbf{x}}' = \tilde{\mathbf{H}}\tilde{\mathbf{x}}$
where H is an arbitary 3×3 matrix



2D Transformations Summary

Transformation	Matrix	DOF	Preserves	Icon
Translation	$[\mathbf{I} \quad t]_{2\times3}$	2	Orientation	
Rigid	$[\mathbf{R} \quad t]_{2\times3}$	3	Lengths	
Similarity	$[sR \quad t]_{2\times3}$	4	Angles	
Affine	$[A]_{2\times3}$	6	Parallelism	
Projective	$[\mathbf{H}]_{3\times3}$	8	Straight Lines	

Order of Transformation

$$T_{4\times4} = \begin{bmatrix} \mathbf{I} & \mathbf{t} \\ \mathbf{0} & 1 \end{bmatrix} \quad R_{4\times4} = \begin{bmatrix} \mathbf{R} & \mathbf{0} \\ \mathbf{0} & 1 \end{bmatrix}$$

$$TR = \begin{bmatrix} \mathbf{I} & \mathbf{t} \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} \mathbf{R} & \mathbf{0} \\ \mathbf{0} & 1 \end{bmatrix} = \begin{bmatrix} \mathbf{R} & \mathbf{t} \\ \mathbf{0} & 1 \end{bmatrix}$$

$$RT = \begin{bmatrix} \mathbf{R} & \mathbf{0} \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} \mathbf{I} & \mathbf{t} \\ \mathbf{0} & 1 \end{bmatrix} = \begin{bmatrix} \mathbf{R} & \mathbf{R}\mathbf{t} \\ \mathbf{0} & 1 \end{bmatrix}$$

$TR = RT$ if: (a) $\mathbf{R} = \mathbf{I}$

(b) $\mathbf{t} = \mathbf{0}$

Thank You